Mokshit Sharma

Software, AI/ML and Data Engineer

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Available for internships and full-time roles, remote, hybrid or on-site in India.

SUMMARY

Founding Engineer & Tech Lead and the only engineer of a live delivery marketplace, building across production full-stack systems, applied machine learning, data pipelines and AI agents that stop for code or a person before the risky step.

SKILLS

Languages	Python, TypeScript, JavaScript, Dart, SQL, Java, C++, R
Backend	Node.js, Express, Python, FastAPI, TypeScript, real-time dispatch, Pusher and Socket.io, Razorpay and PayPal payments, ledgers and idempotency, scheduled jobs
Frontend and mobile	Flutter (Play Store and App Store releases), React, Next.js, React Native (Expo), Tailwind CSS
Databases	MongoDB (geospatial), PostgreSQL (Prisma, Drizzle), Supabase, MariaDB, SQLite, Chroma, Redis
Agents and LLMs	LangGraph (interrupts, checkpoints), Model Context Protocol servers and clients, tool calling and planning, human approval in the loop, structured output (Zod, Pydantic), RAG with citations, Groq, Gemini and Claude APIs
Real-time voice	Streaming speech-to-text and text-to-speech (Deepgram, faster-whisper), WebSocket audio, barge-in
Machine learning	LightGBM, XGBoost, CatBoost, scikit-learn pipelines, PyTorch (LSTM, temporal CNN), Stable-Baselines3 (PPO), hidden Markov models, SHAP, calibration, leakage-safe validation
NLP and vision	FinBERT, TF-IDF, VADER, OpenCV
Data and BI	SQL (joins, CTEs, window functions), pandas, NumPy, Power BI, Excel, ETL pipelines, Parquet, R (tidyverse)
Algorithms	Java, hand-built hash map, heap and k-d tree, Dijkstra, A*, 2-opt, C and C++
Delivery	Docker, GitHub Actions, Codemagic, Vercel, pytest, Vitest, JUnit, Playwright

EXPERIENCE

Founding Engineer & Tech Lead, Movigo Innovations Pvt. Ltd., Indore Apr 2026 – present

App Developer (Apr 2026), Lead App Developer (May 2026), Founding Engineer & Tech Lead (Jun 2026)

- Only engineer at a live delivery marketplace: rebuilt the product from scratch after an outsourced team delivered designs but no working software, and own the backend, both mobile apps, the operations console and production support.
- Node.js, Express and MongoDB backend with 400+ API routes, 40 data models, 8 scheduled jobs and a 15-second real-time dispatch loop.
- Made booking acceptance race-safe: one conditional database write lets exactly one of several accepting drivers win a booking.
- Shipped Flutter driver and retailer apps to Google Play (1K+ installs each) and the retailer app to the App Store.
- Moved pricing, zones, surge multipliers and dispatch radius into cached server-side settings: operations change prices without an app release, and pricing reads fall from 100+ per second to none within 60 seconds.
- Switched dispatch to OSRM road distances with one distance-matrix call per booking instead of one call per driver.
- Integrated Razorpay wallets and fees so every payment is verified and credited exactly once; moved live tracking and chat to Pusher without breaking apps already in users' hands.
- Built FieldOps, a workforce platform for a ~20-person field team: GPS attendance from an Android foreground service uploaded in one batch a minute, onboarding, call logs and a points engine with weights stored in the database.

Internships

- **Data Science Intern**, Kangaroo Software Pvt. Ltd., Indore Oct 2025 – Jan 2026. Worked on live project modules.
- **Data Science Intern (virtual)**, CodeAlpha Nov – Dec 2025
- **Power BI Intern (virtual)**, SaiKet Systems Sep – Oct 2025. Data analysis and visualisation in Power BI.
- **AI/ML Intern and Python Development Intern (virtual)**, CODEXINTERN Aug 2025
- **Software Development Engineer Intern (part-time, remote)**, Bluestock Fintech May – Jun 2025
- **Data Analyst Intern**, Afame Technologies Feb – Mar 2025
- **C and C++ Programming Intern**, TechnoHacks Solutions Pvt. Ltd. Nov – Dec 2024
- **C++ Programming Intern (virtual)**, CodSoft Nov – Dec 2024
- **Artificial Intelligence Intern**, Evoastra Ventures Oct – Nov 2024
- **Data Science Intern**, Cognifyz Technologies Sep – Oct 2024. Collected, cleaned and interpreted datasets.
- **Data Science and Machine Learning Internship Program**, Edureka Jun 2024

PROJECTS

SettleAI Next.js, TypeScript, PostgreSQL, Drizzle, MCP, PayPal APIs, Zod, Playwright PayPal AI Hackathon 2026 entry

- Escrow for real-world jobs that AI agents hire people to do: agents scope jobs, judge photo proof and propose dispute splits, and a deterministic guard approves every payout.
- Double-entry ledger in integer cents with idempotency keys; every step becomes a signed, hash-linked receipt that verifies offline in 81 ms.
- 146 unit, integration and end-to-end tests; evaluation shows 0% false releases and every injected or tampered proof detected, and a 50-way concurrency test never overspends.

ToolForge Python, LangGraph, MCP, Pydantic, Typer, pytest

- Agents turn an OpenAPI spec into an MCP server, test it against the live API and audit it for security; a person approves the tool plan through a LangGraph interrupt.
- Pass or fail comes from recorded API calls: the PokéAPI build passed 8 of 8 live calls with a 100/100 security score; a 102-endpoint spec fits a 16,000-character prompt budget for free-tier models; 24 pytest tests.

Naman, voice agent Python, LangGraph, Groq, Gemini, MCP, faster-whisper, SQLite, Telegram

- Voice agent for laptop and phone with 66 tools in 12 skill packs; approval is its own graph node, so every tool runs exactly once after a pause; 164 pytest tests.
- A code verifier cut outreach emails with an unsupported claim from 48 of 49 to 0 of 21; templated emails use zero AI tokens, and a simple command sends 22 of 66 tool schemas instead of all.
- Scheduled job-search pipeline across LinkedIn and Indian job boards found 1,555 openings by October 2026, with 30-day caching, pre-fetch filtering and quota-aware fallback across models.

VoiceDesk Node.js, WebSocket, AudioWorklet, Deepgram, Groq, MongoDB github.com/Mokshitsharma/VoiceDesk---AI-Voice-calling-agent

- Real-time voice receptionist with streaming speech-to-text and speech: books appointments from spoken dates, hands off to a person, and stops talking the moment the caller interrupts.
- Rule-based fast paths answer common requests with no model call; every turn is measured against a 1.5-second round-trip budget.

docIQ Node.js, Express, MongoDB, Chroma, BullMQ, Next.js, Gemini, Groq

- Question answering over contracts and invoices with answers that cite their source passage, plus a clause risk scan.
- Hybrid BM25 and vector retrieval; offline evaluation 11 of 11 retrieval hits and 1 of 1 correct refusal; 16-check end-to-end suite; falls back to extractive answers without an API key.

CricXAI Python, LightGBM, SHAP, FastAPI, Parquet, Docker, GitHub Actions github.com/Mokshitsharma/CricXAI

- Recommends what to bowl next in an ODI, with dismissal probability, dismissal type, expected runs and SHAP reasons, trained on 1.36 million real deliveries.
- Leakage-safe features, match-grouped cross-validation and isotonic calibration: dismissal ROC-AUC 0.749, runs MAE 0.867; versioned models served through FastAPI.
- Parquet store about 15× smaller and 10× faster to load than CSV; 71 pytest tests with CI on GitHub Actions.

Sensei AI Python, PyTorch, Stable-Baselines3, hmmlearn, SHAP, FinBERT, FastAPI, Next.js github.com/Mokshitsharma/Sensei

- Buy, sell or hold signals for Nifty 50 stocks from five model families (random forest, LSTM, temporal CNN, PPO agent, hidden Markov regime model) combined in a readable vote with SHAP and FinBERT news sentiment.
- Backtester reports returns, Sharpe ratio and maximum drawdown; caching loaded models cut a cold analysis from about 31 s to about 7 s.

AI Business Analytics Python, scikit-learn, SHAP, Gemini, FastAPI, Streamlit, ReportLab github.com/Mokshitsharma/ai-business-analytics

- Upload a CSV and get cleaning, exploratory analysis, automatically trained models, SHAP explanations and an LLM-written HTML and PDF report.
- One pipeline behind a CLI, a FastAPI service with background jobs and a Streamlit app; SHAP cost bounded by a 100-row sample; the report still ships if the LLM is unavailable.

Customer churn with explainable AI Python, XGBoost, SHAP, Streamlit github.com/Mokshitsharma/Customer_Churn_using_XAI

- Predicts which customers will leave with XGBoost and explains each prediction in plain English, with a retention suggestion, using SHAP; Streamlit dashboard for non-technical users.

RouteCore Java 17, Maven, JUnit; no runtime dependencies github.com/Mokshitsharma/RouteCore

- Route-optimisation and fleet-scheduling engine with a hand-written hash map, min-heap and k-d tree, Dijkstra, A* and 2-opt.
- k-d tree 408× faster than brute force at 100,000 points (3.92 ms against 1,601.66 ms); A* expands up to 200× fewer nodes; 47 JUnit tests.

Refera React Native (Expo), React, Express 5, Prisma, PostgreSQL, TypeScript github.com/Mokshitsharma/Banksaathi-clone

- Fintech referral platform: converted leads pay two-tier commissions into an append-only ledger where a database trigger rejects updates and deletes and balances can never go negative.
- Money in integer paise and rates in basis points; row locks make simultaneous withdrawals safe; mobile app, admin panel and API in one typed monorepo with 50 API tests.

Hyperlocal Marketplace Node.js, Express, PostgreSQL, Drizzle, Socket.io, Next.js, TypeScript

- Delivery marketplace with a customer web app, store dashboard and operations API: every order transition is a compare-and-set update, stock is reserved atomically, and in-flight orders are reconciled on restart.
- 52 of 52 backend tests passing, covering the state machine, checkout and payments, dispatch retry and escalation, and restart recovery.

Dairy Collection Manager React, Vite, Supabase, PostgreSQL, Vercel Live · milk-management-five.vercel.app

- Daily milk register for a collection centre: fat-based rate interpolation, supplier ledgers, advances repaid oldest-first and period settlements in one reversible database transaction.

Paytm stock dashboard Python, yfinance, Power BI

- Four-stage Python ETL aligning Paytm's share price with competitors and market events, feeding two Power BI dashboards.

NIFTY 50 dashboard Python, pandas, yfinance, Power BI github.com/Mokshitsharma/nifty-50-dashboard-using-python-and-power-bi

- Fundamentals and price history for all 50 constituents pulled with yfinance and pandas and modelled in Power BI.

Kaggle Playground competitions Python, scikit-learn, LightGBM, CatBoost, XGBoost

- Diabetes prediction over 26 health features with a blended CatBoost and LightGBM model; LightGBM with early stopping reached a validation AUC of 0.728.
- Student-score regression with K-fold validation and a stacked HistGradientBoosting and XGBoost ensemble under a ridge meta-model.

EDUCATION

Integrated B.Tech + M.Tech, Artificial Intelligence and Data Science, CGPA 8.2 2023 – 2028 (expected)
School of Data Science and Forecasting, Devi Ahilya Vishwavidyalaya, Indore

CERTIFICATIONS

Google Data Analytics Professional Certificate; HackerRank 5-star in Python, SQL and C++; Data Analyst Program, Infosys Springboard; 5-Day AI Agents Intensive, Kaggle and Google (Dec 2025); Generative AI Foundations, Skillsoft (Nov 2025); Generative AI Foundations: IT Integration with Generative AI, Infosys Springboard (Nov 2024); Excel, Infosys Springboard (Nov 2024); AI Masterclass, Freedom of AI (Jan 2024).